

ELIZABETH CITY, NC, USA

WMO: 723070

Lat: 36.261N

Lon: 76.175W

Elev: 4

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13786

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.9	(c) -3.7	(d) -14.6	(e) 1.1	(f) -3.1	(g) -12.2	(h) 1.3	(i) -1.1	(j) 12.0	(k) 3.5	(l) 11.1	(m) 5.3	(n) 4.3	(o) 290	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	33.8	25.7	32.6	25.3	31.4	24.7	26.9	31.5	26.4	30.5	25.7	29.6	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.2	25.2	20.3	28.6	24.6	19.6	28.0	84.7	31.5	81.7	30.4	79.1	29.6	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.9	(b) 9.2	(c) 8.3	(e) -9.5	(f) 35.7	(g) 2.4	(h) 1.4	(i) -11.2	(j) 36.7	(k) -12.6	(l) 37.5	(m) -14.0	(n) 38.3	(o) -15.7	(p) 39.4		
(a) 10.9	(b) 9.2	(c) 8.3	(e) -10.6	(f) 28.2	(g) 2.3	(h) 1.0	(i) -12.2	(j) 28.9	(k) -13.5	(l) 29.4	(m) -14.8	(n) 30.0	(o) -16.4	(p) 30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.7	7.3	10.5	15.8	20.5	24.7	26.4	25.9	23.2	17.6	11.9	8.2
	DBStd	8.49	5.86	5.12	5.17	4.63	3.96	2.84	2.28	2.17	2.90	4.46	4.66	5.33
	HDD10.0	463	159	106	58	6	0	0	0	0	0	2	32	101
	HDD18.3	1665	392	311	248	103	25	1	0	0	3	68	198	317
	CDD10.0	2847	26	29	74	179	326	442	508	494	396	239	89	44
	CDD18.3	1006	1	1	5	27	93	193	249	236	148	46	5	2
	CDH23.3	8546	2	6	34	198	693	1704	2456	2148	1019	264	18	4
	CDH26.7	2861	0	0	3	40	194	620	939	747	265	52	1	0
Wind	WSAvg	4.0	4.3	4.5	4.7	4.8	4.2	3.8	3.5	3.3	3.8	3.8	3.9	3.9
Precipitation	PrecAvg	1214	100	86	97	76	104	111	144	135	114	88	74	80
	PrecMax	1842	274	179	209	161	264	223	355	370	443	236	212	244
	PrecMin	916	32	25	26	14	14	30	21	30	4	0	14	8
	PrecStd	197	47	40	46	38	54	49	76	72	83	64	42	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.3	26.8	29.9	32.8	35.2	35.6	34.4	32.6	30.1	25.5	23.7
		MCWB	17.9	18.6	18.1	21.3	24.1	24.8	26.8	26.1	24.9	22.8	20.5	19.0
	2%	DB	19.7	21.0	23.8	27.7	30.5	33.2	33.8	32.9	31.0	28.1	23.1	21.0
		MCWB	16.5	15.8	17.0	19.8	22.8	24.5	26.4	25.7	24.2	22.0	18.7	18.3
	5%	DB	17.5	18.6	21.6	25.8	28.9	31.7	32.5	31.8	29.5	26.3	21.3	19.0
		MCWB	15.0	15.2	15.8	18.9	22.1	24.3	25.8	25.4	23.6	21.0	17.6	16.7
	10%	DB	15.1	16.2	19.1	23.5	27.2	30.2	31.2	30.6	28.1	24.4	19.4	17.2
		MCWB	12.5	12.5	15.2	17.9	21.2	23.6	25.2	24.7	23.0	20.3	16.6	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.4	19.8	23.2	25.3	26.8	28.3	27.7	26.6	25.0	22.1	20.5
		MCDB	21.4	22.7	24.2	27.3	29.9	31.4	33.3	32.6	30.1	27.3	23.6	22.1
	2%	WB	17.3	17.6	18.5	21.3	24.1	25.8	27.3	26.8	25.7	23.6	20.0	18.7
		MCDB	19.2	20.0	22.3	25.5	28.3	30.4	32.3	30.9	28.7	26.3	22.1	20.5
	5%	WB	15.2	15.5	17.2	20.1	23.1	25.2	26.6	26.1	24.9	22.5	18.6	17.2
		MCDB	17.1	18.1	20.2	24.2	27.0	29.8	31.2	29.8	27.9	25.1	20.5	18.8
	10%	WB	13.0	12.7	15.6	18.9	22.2	24.6	25.8	25.6	24.2	21.3	16.9	15.0
		MCDB	14.7	15.6	18.9	22.6	26.1	28.8	29.9	29.1	26.9	23.6	19.0	16.8
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.3	10.6	10.7	9.8	9.2	8.6	8.4	8.5	10.1	10.6	9.9
		MCDBR	12.5	13.3	13.4	12.9	11.1	10.7	9.9	9.6	9.8	10.9	11.6	11.0
		MCWBR	9.6	9.5	7.8	6.5	5.1	4.2	4.3	3.9	4.3	5.6	7.5	8.6
	5% WB	MCDBR	11.7	12.4	11.4	11.1	10.0	9.3	9.1	8.7	7.8	8.5	9.8	10.1
		MCWBR	10.0	9.8	8.2	6.6	5.0	4.2	4.3	3.9	3.9	5.0	7.6	8.8
Clear-Sky Solar Irradiance	taub	0.312	0.325	0.351	0.394	0.438	0.490	0.516	0.505	0.413	0.361	0.328	0.318	
	taud	2.535	2.492	2.448	2.337	2.245	2.152	2.090	2.109	2.370	2.499	2.538	2.549	
	Ebn at Noon	889	915	918	890	850	804	779	779	842	863	861	858	
	Edh at Noon	79	92	106	124	138	152	161	155	113	91	78	74	
All-Sky Solar Radiation	RadAvg	2.41	3.17	4.21	5.38	6.02	6.38	6.05	5.47	4.50	3.62	2.71	2.15	
	RadStd	0.18	0.33	0.32	0.39	0.46	0.36	0.30	0.34	0.32	0.40	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+231	+163
(47) Regional (53 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+112

Nomenclature: See separate page